Privacy-Preserving Network Security: Classifying Encrypted Malicious Traffic and Handling Class Imbalance using Ensemble Machine Learning

Moeez Jamal (Roll No: 22i0513)

Ammad Nasir (Roll No: 22i0477)

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Abstract

The rapid proliferation of sophisticated cyber threats has highlighted critical limitations in traditional, signature-based Network Intrusion Detection Systems (NIDS). More importantly, with over 95% of modern internet traffic now encrypted via TLS/SSL, legacy Deep Packet Inspection (DPI) is becoming functionally obsolete. While machine learning offers a powerful mechanism to classify encrypted threats purely on behavioral metadata (without breaking user privacy), this transition introduces a severe structural flaw: extreme class imbalance. Benign traffic flows continually outnumber malicious attack packets, heavily skewing conventional predictive models and leading to high false-negative rates for critical minority classes like zero-day exploits or C2 botnets. This research proposes the development of a state-of-the-art, developable machine learning pipeline engineered to classify highly imbalanced, encrypted network traffic. By leveraging the comprehensive CICIDS2017 dataset, this project will evaluate the intersection of hybrid resampling techniques (e.g., SMOTE combined with Tomek Links) and cost-sensitive ensemble algorithms, primarily XGBoost. The primary objective is to maximize the Precision-Recall Area Under Curve (PR-AUC) on minority encrypted attacks without decrypting payloads. Outcomes from this research will yield a fully trained, deployable classification model capable of detecting minute anomalous behaviors in encrypted tunnels, directly solving the modern enterprise dilemma of balancing user privacy with robust network defense.

1. Introduction

“The single biggest threat to any digital infrastructure is the traffic we fail to categorize.” In today’s hyper-connected landscape, where vast terabytes of data flow dynamically across decentralized and cloud-native architectures, securing network boundaries requires more than static firewalls. Network traffic classification remains the cornerstone of network management and anomaly-based intrusion detection. However, machine learning’s integration into this domain introduces a significant structural obstacle: the sheer ubiquity of standard, normal traffic versus the relatively minuscule footprint of malicious probing, infiltration, or botnet behaviors. Because algorithms naturally strive for high global accuracy, they frequently ignore these statistically rare—yet critically damaging—events, producing a dangerous false sense of security.

As undergraduate researchers with prior exposure to data structures, advanced programming, and foundational data science concepts, we recognize that the future of Computer Networks relies heavily on automated, adaptive protocols. Building a model that simply categorizes 99% of benign traffic accurately is functionally inadequate if it entirely misses the 1% of traffic representing a lethal ransomware payload. Therefore, this project pivots from standard accuracy paradigms to focus aggressively on methodologies specifically tailored for skewed arrays, reflecting the harsh realities of real-world packet captures.

The relevance of imbalanced learning in contemporary cybersecurity cannot be overstated, particularly due to the "Encryption Dilemma." Modern enterprise systems face adversaries who mask command-and-control (C2) communications within standard HTTPS overhead. Because privacy laws and Zero-Trust architectures prohibit the decryption of payloads for deep inspection, security must rely entirely on the "shape" of the traffic—flow durations, packet size variations, and inter-arrival times. When algorithms process this encrypted metadata, the overwhelming volume of benign web scraping effectively blinds standard supervised learning classifiers to the anomalies. To detect malware in encrypted tunnels without breaching privacy, standard algorithmic behavior must be meticulously audited and fundamentally altered.

In this research, we aim to design and implement an end-to-end predictive pipeline that actively fights back against majority class dominance. We anchor our effort on scalable, tree-based ensemble methods—specifically XGBoost and Random Forest architectures—due to their intrinsic capability to partition non-linear boundaries alongside natively implemented cost-sensitive learning functions. By introducing synthetic minority oversampling (SMOTE) and combining it with undersampling links, our objective is to synthesize a robust boundary layer. This project not only bridges our academic coursework in computer networks and machine learning but also results in a tangible, highly developable artifact: a streamlined model optimized for practical deployment in packet-sniffing software like Wireshark or Suricata, ready for the industry's rigorous standards.

2. Problem Statement

Network Intrusion Detection Systems (NIDS) persistently suffer from two compounding failures: (1) legacy Deep Packet Inspection is blind to encrypted payloads, meaning detection must rely on raw flow statistics, and (2) datasets mapping these statistics suffer from extreme class imbalance, severely degrading their ability to classify minority attack vectors. Conventional machine learning models default to majority-class biases, inflating false-negative rates and allowing encrypted threats to slip through undetected. This proposal targets cybersecurity engineers and network administrators tasked with monitoring modern, privacy-first enterprise traffic. Over the mandated project timeline, we aim to address a core challenge: How can we computationally redefine the decision boundaries of network classifiers so that mathematically rare, encrypted cyber-attacks are categorized with the exact same programmatic urgency as high-volume benign traffic, entirely without decrypting the data?

3. Objectives

To systematically fulfill the problem statement, the project outlines four declarative objectives:

1. To investigate the detrimental impact of data imbalance on the predictive capabilities of algorithms deployed specifically on encrypted network flows (metadata vs. payload).

2. To synthesize and implement optimized hybrid sampling techniques—specifically combining SMOTE with Tomek Links—tailored for handling the high-dimensionality of encrypted packet metadata.

3. To evaluate and iterate upon cost-sensitive ensemble architectures, specifically comparing XGBoost and Random Forest models, focusing explicitly on increasing minority-class retention rates.

4. To deliver a functional, highly developable classification pipeline that minimizes both computational latency and false negatives, ensuring applicability for real-time network traffic analysis.

4. Research Questions and Hypotheses

Our study is guided by the following essential inquiries and academic propositions:

- **RQ1:** How does the application of hybrid resampling techniques (e.g., SMOTE-Tomek) quantitatively adjust the precision-recall trade-off when compared to traditional, un-sampled baselines on network data?
- **RQ2:** Which supervised ensemble modeling approach retains the highest operational robustness while processing massively imbalanced, non-linear IP packet descriptors?
- **H1:** We hypothesize that employing a cost-sensitive variant of XGBoost, supplemented with intelligent minority over-sampling, will yield a significantly higher F1-score and PR-AUC than standardized multi-class decision boundaries.

5. Dataset Description

For our computational environment, we will utilize the Canadian Institute for Cybersecurity's **CICIDS2017 Dataset**, universally renowned for its accurate reflection of complex, modern network attacks combined with diverse background traffic.

- **Source and Scale:** Available openly from the University of New Brunswick (UNB), encompassing over 2.8 million structured records representing 5 days of captured network traffic.
- **Type and Features:** It provides purely tabular, structured data with 78 multi-dimensional behavioral features extracted using CICFlowMeter (e.g., Flow Duration, Packet Length Variables, Inter-Arrival Times).
- **Imbalance Ratio:** The dataset natively highlights severe disparities: nearly 80% benign behavior opposed to minuscule fractional traces of Web Attacks, Infiltration, or Heartbleed attacks.

- **Ethical Considerations:** Built on controlled lab environments, the dataset eliminates privacy concerns regarding personally identifiable information (PII) typically located in raw, un-anonymized ‘.pcap‘ files.

6. Methodology

The practical evolution of our model demands a rigorous, repeatable pipeline mapped directly to industry-standard data mining frameworks:

6.1 Data Understanding and EDA

Initial processing requires addressing infinite values and empty data inputs inherent to parsed network features. We will conduct deep Exploratory Data Analysis (EDA)—visualizing extreme variations in protocol distributions (TCP/UDP) alongside kernel density estimations to fully map the class skewness defining the anomaly vectors.

6.2 Preprocessing and Feature Engineering

To prepare the dataset for algorithmic ingestion, we will undergo significant feature engineering. Highly correlated or zero-variance elements will be dropped to improve processing speed. Categorical data (like IP representations or string-based Port mappings) will be properly encoded, and numeric variables will undergo robust scaling to normalize the wide variation in bytes-per-second statistics.

6.3 Baseline Modeling

We establish the control group using shallow logic (Logistic Regression, Naive Bayes) alongside un-tuned Random Forest iterations. This demonstrates the exact numeric vulnerability the system faces prior to sophisticated handling, ensuring all resulting optimizations are measurably validated.

6.4 Imbalance Handling and Optimization

At the core of the study, the system tackles class disparity. We will deploy *SMOTE* to synthetic network data mathematically mirroring rare attacks. Simultaneously, *Tomek Links* algorithms will undersample borderline cases between benign overlap and attack states to clean decision boundaries. To prevent memory exhaustion during development, cost-sensitive (class weight manipulation) techniques will be deeply integrated directly within model hyperparameters.

6.5 Validation and Robustness

Stratified K-Fold cross-validation ensures train-test splits proportionally represent the rarest hacking classes. To test for robustness simulating a real operational environment, models will additionally be evaluated on independent, temporal slices of the dataset to check for conceptual drift across different days of the week.

6.6 Reporting and Visualization

Final results will be collated mathematically and visually. Precision-Recall curves (superior to standard ROC on imbalanced sets), normalized confusion matrices, and SHAP (SHapley Additive exPlanations) values will determine feature importance, illustrating geographically *why* specific network packets triggered classification flags.

7. Tools and Technologies

To ensure the pipeline is highly developable, repeatable, and leverages the latest optimizations, the technical stack includes:

- **Core Engine:** Python 3.10+
- **Data Handlers:** Pandas and NumPy for computationally lightweight multidimensional array structuring.
- **Machine Learning & Imbalance Sets:** Scikit-learn, Imbalanced-learn (for SMOTE/Tomek functions), and XGBoost.

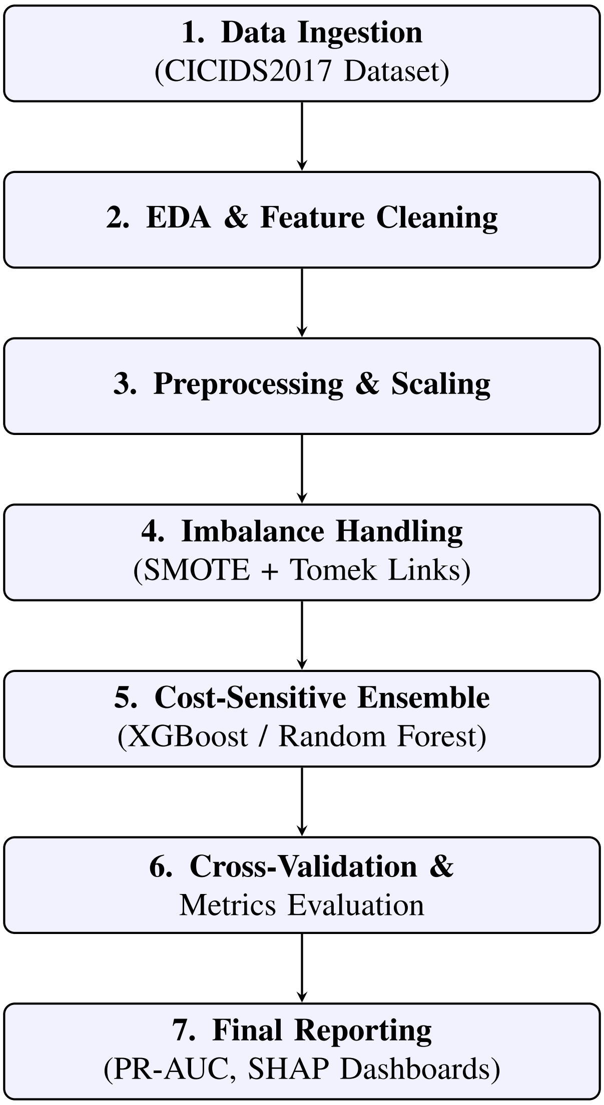

Figure 1: Architecture and Stepwise Workflow of the NIDS Classification Pipeline

- **Visualization:** Matplotlib and Seaborn for high-fidelity confusion matrices and correlation heatmaps.
- **Environment:** Google Colab Pro or standard Jupyter Lab ecosystems for rigorous GPU execution.

8. Expected Outcomes

Our delivery will manifest in four primary artifacts: 1. A fully developable, optimized ensemble classification model coded in Python explicitly configured for imbalanced packet categorization. 2. A formalized, comparative data analysis report reflecting how various permutations of synthetic sampling modify baseline accuracies. 3. An IEEE computationally formatted final technical report outlining exact experimental structures. 4. Pythonic dashboards or notebook-based visualizations clearly denoting feature importance.

9. Evaluation Metrics

Because traditional Accuracy is deeply flawed when reviewing skewed distributions (e.g., 99% benign traffic yields 99% accuracy by predicting everything as benign), performance quantification relies strictly on:

- **F1-Score (Macro and Micro):** Unifying Precision and Recall.
- **Precision-Recall AUC (PR-AUC):** The defining metric for tracking improvement in isolated minority performance.
- **Recall (Sensitivity):** Specifically monitored to trace how aggressively the model captures attacks without marking false negatives.

References

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